

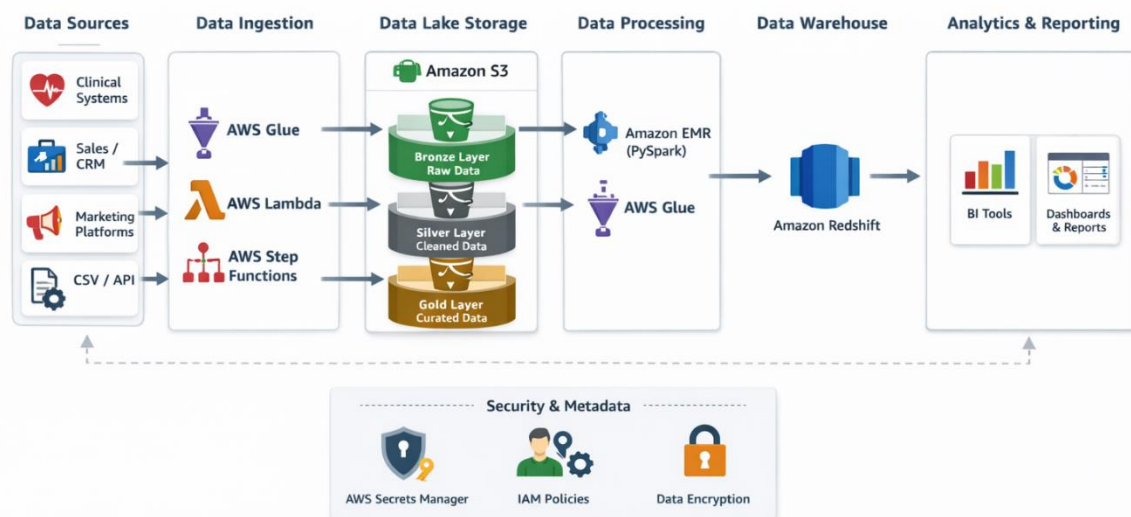
Project Title

Pharma Commercial Data Platform – AWS Lakehouse for Business Analytics

Business Context

The pharmaceutical client operated across multiple domains including clinical research, sales distribution, and marketing campaigns. Data was generated from various systems but lacked integration, making it difficult to derive meaningful business insights.

Architecture



Key Challenges

- Disparate data sources (clinical, sales, marketing)
- No unified data model for analytics
- High dependency on manual reporting
- Data quality inconsistencies across systems
- Performance issues with large-scale data processing

Technology Stack

- AWS Glue – Designed and orchestrated data ingestion and transformation pipelines
- Amazon S3 – Implemented layered storage architecture (Bronze, Silver, Gold)
- Amazon EMR – Performed large-scale distributed data processing using PySpark
- Amazon Redshift – Enabled analytical querying and reporting
- AWS Lambda – Handled event-driven data processing
- AWS Step Functions – Managed end-to-end pipeline orchestration
- PySpark – Data transformation and enrichment
- SQL – Data querying and validation
- AWS Secrets Manager – Managed secure access to credentials

Solution Approach

Designed a scalable **AWS-based Lakehouse architecture** to centralize and process data efficiently.

- Built ingestion pipelines using AWS Glue and event-based triggers
- Stored data in Amazon S3 using layered architecture
- Processed large datasets using Amazon EMR (PySpark)
- Created curated datasets and exposed them via Amazon Redshift

Solution Implementation

◆ Data Ingestion

- Batch and incremental ingestion pipelines
- Metadata-driven ingestion design
- Handled schema drift at ingestion layer

◆ Data Processing

- Data cleaning, transformation, and enrichment
- Standardized schemas across datasets
- Integrated multiple data sources (sales + product + region)

◆ Data Modelling

- Designed **star schema** with tables
- Fact tables: sales, prescriptions, campaign performance
- Dimension tables: product, doctor, region
- Optimized for analytical queries

◆ Data Storage Strategy

- Bronze: Raw data storage
- Silver: Cleaned and validated data
- Gold: Aggregated and business-ready datasets

Advanced Engineering Practices

- Implemented incremental processing using watermark logic
- Designed idempotent pipelines for safe reprocessing
- Applied partitioning strategy for large datasets
- Optimized Spark jobs using caching and broadcast joins
- Implemented logging and monitoring for pipeline tracking

Data Governance & Security

- IAM-based access control
- Secure credential management using AWS Secrets Manager
- Data encryption at rest and in transit
- Controlled access at S3 bucket level

Outcome / Business Impact

- Centralized pharma data into a unified platform
- Reduced manual reporting effort significantly
- Enabled faster insights into drug performance
- Improved decision-making with reliable data
- Scalable architecture capable of handling large datasets